Differential Equations

1. The Pendulum
   - a first mathematical model
   - solving with Laplace transforms
   - the slope field
2. The Pendulum with Damping
   - a second mathematical model

MCS 320 Lecture 31
Introduction to Symbolic Computation
Jan Verschelde, 24 July 2026

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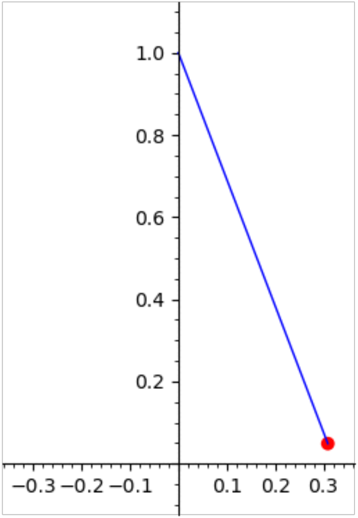
The Pendulum



An ordinary differential equation with initial conditions, e.g.:

$$L \frac{d^2}{d\,t^2} \theta(t) = -g\,\theta(t),$$

$$\theta(0) = \pi/10, \quad \theta'(0) = 0,$$

is an *initial value problem*.

In the model of the pendulum, $\theta(t)$ is the angle of deviation from the equilibrium position, as a function of time t.

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Laplace Transforms

The Laplace transform of $L\dfrac{d^2}{d\,t^2}\theta(t) = -g\theta(t)$
is
$$\left(\Theta(s)s^2 - \theta(0)s - \theta'(0)\right) L = -g\Theta(s), \quad \Theta(s) = \mathcal{L}(\theta(t)).$$

The transformed equation is linear in $\Theta(s)$, its solution is

$$\Theta(s) = \frac{Ls\theta(0) + L\theta'(0)}{Ls^2 + g}.$$

Substituting $\theta(0) = \pi/10$ and $\theta'(0) = 0$ gives $\pi/10Ls/(Ls^2+g)$.

The inverse Laplace transform of $\Theta(s)$ is

$$\theta(t) = \frac{\pi}{10}\cos\left(\frac{t}{L}\sqrt{Lg}\right).$$

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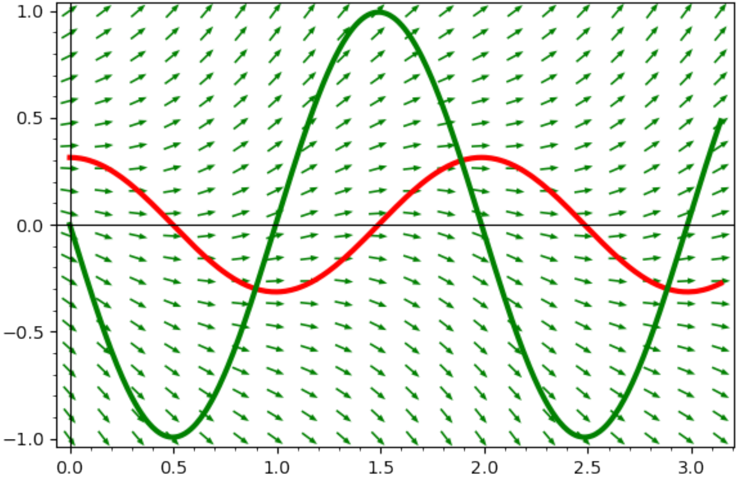
The Slope Field for $g = 10$ and $L = 1$



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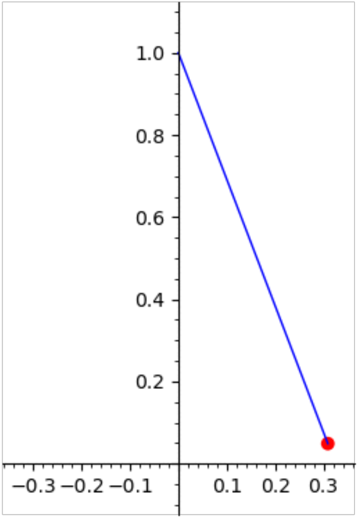
The Pendulum with Damping



A second model of the pendulum includes damping, and $\sin(\theta(t))$:

$$L\frac{d^2}{dt^2}\theta(t) = -A\frac{d}{dt}\theta(t) - g\sin(\theta(t)),$$

$$\theta(0) = \pi/10, \quad \theta'(0) = 0.$$

Damping is proportional to the velocity.

In the model of the pendulum, $\theta(t)$ is the angle of deviation from the equilibrium position, as a function of time t.